# PRIME FACTOR STUDIO

Multiply by Stacking: Tactile Prime Towers & 12-Socket Lid Tray



**WHAT IS THIS BOX FOR? MEET THE BLOCKS THAT MULTIPLY WHEN YOU STACK THEM.**

Most counting blocks add (2 + 3 = 5). Ours use a secret logarithmic superpower: stack [ 2 ] on [ 3 ] and the tower height multiplies to 6!

## THE SECRET SUPERPOWER: MULTIPLYING BY HEIGHT

Every whole number is built from indestructible "prime atoms" (2, 3, 5, 7...).
Because each block's height is scaled to the logarithm of its prime:

**Height(a × b) = Height(a) + Height(b)**

Whenever you snap blocks together into a tower, adding their physical heights automatically multiplies their numbers — no calculator required!

### 12-SOCKET COMPARISON LID TRAY

Place the 12-socket lid tray flat on your desk to hold prime towers upright side-by-side! Compare quantities and factorizations visually — whenever two towers stand at the exact same height on the lid tray, their products match.

### WHAT'S INSIDE YOUR BOX (72 PIECES)

- 70 Prime Blocks: 29×[2], 17×[3], 8×[5], 4×[7], 3×[11], 3×[13], 2×[17], 2×[19], 2×[23]
- 1× Storage Box with Sliding Door + 1× 12-Socket Comparison Lid Tray
- Keep this folded guide inside your box lid for examples & part IDs!

**★ YOUR FIRST 60 SECONDS WITH THIS BOX** — START HERE!

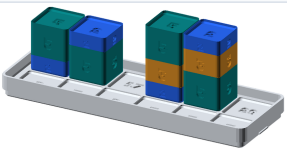
1. **SLIDE OPEN THE BOX DOOR & POP THE LID:**
   Slide the front door up and place the 12-socket Lid Tray flat on your desk as your stage.
2. **COMPARE TWO TOWERS ON THE LID TRAY:**
   Stand [2]+[3] (= 6) in one socket and [2]+[2]+[3] (= 12) next to it to see the height difference!
3. **PRESS THE BOX ICON KEY [g][FACT]:**
   Type 12 [g] [FACT] on your calculator—the factor tree icon on your box lid is this exact key!

**BOX ICON = CALCULATOR [PRIME] & [G][FACT] FACTOR TREE KEY**

The button icon embossed on your box lid (left) is the exact calculator key ([PRIME] / [g][FACT])! Build every lesson with your blocks, then press the key.

## COMMUTATIVE TOWERS: [2]×[3]×[5] = 30 ↔ BOX [PRIME] ICON

**STEP A: HANDS-ON WITH BLOCKS ON LID TRAY** — 1:1 CAD SCENE



| Tower [2][5] = 10 | Tower [5][2] = 10 | Tower [2][3][5] |
|---|---|---|
| 23.25 mm | 23.25 mm | 34.34 mm (= 30) |

▲ **Stack prime blocks in any order on the Lid Tray: both 10-towers and both 30-towers stand dead-level!**

**STEP B: TRY IT ON YOUR CALCULATOR** — BOX ICON: [PRIME] / [g][FACT

FACTOR TREE: 30 = 2 × 3 × 5 (34.34 MM)

30 (34.34 mm)

6 (18.09 mm) × 5

[2] Blue [3] Gold [5] White

7.00 + 11.09 + 16.25 = 34.34 mm

PRESS BOX ICON & RPM KEYS (SCG):

1. STACK 2×3×5:

2 ENTER 3 × 5 ×

2. BOX ICON KEY:

g FACT 2×3×5 = 30

✓ DISPLAY (X) = 30

**KEYSTROKES:** 2 [ENTER] 3 [×] 5 [×] [g] [FACT]
Stack [2], [3], and [5] on the Lid Tray (= 30), then press [g][FACT] to see the prime tree!

# HOW IT WORKS: THE CORE IDEA

Logarithmic Height Scaling: Stack to Multiply



## THE LOGARITHMIC STACKING PRINCIPLE

Because logarithms turn multiplication into simple addition, every prime block has a visible height proportional to log10(p):

**h_vis(p) = K_log × log10(p)**

When you snap blocks together, the top socket completely swallows the bottom plug (1.85mm), so only the visible heights add up:

**H_tower(N) = 1.85 mm + K_log × log10(N)**

### HOW THE SCALE IS TUNED

- Base Unit Height: Prime [ 2 ] = 7.00 mm (matches our 1/12 Fraction block!)
- Logarithmic Scale Factor: K_log = 7.00 / log10(2) ≈ 23.25 mm per factor of 10
- Flush Snap Connector: 1.85 mm plug seats cleanly into 1.85 mm socket
- Unique Prime Fingerprint: Only one combination of primes reaches each height!

### COMMUTATIVE PROPERTY: A × B = B × A

In math class, you're told "order doesn't matter when multiplying." Here, you can see why! Build [2]→[2]→[3]→[5] (= 60) and [5]→[3]→[2]→[2] (= 60). Seat them side-by-side on the lid tray: both towers are 41.34 mm tall. Rearranging the blocks never changes the total height of the tower!

### 3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE

VISUAL PROOF

✗ **TRAP 1 — Confusing $2^3$ with 2 × 3:**
✓ **FIX:** On the Lid Tray, [2]·[2]·[2] (= 8, 21mm) stands visibly taller than [2]+[3] (= 5, 18mm)!

✗ **TRAP 2 — Mixing up GCD and LCM:**
✓ **FIX:** On the Lid Tray, GCD is the shorter shared tower (6); LCM is the taller union tower (36).

✗ **TRAP 3 — Thinking 1 is a prime:**
✓ **FIX:** A '1' block would have height log(1) = 0 mm—it wouldn't change any tower's height at all!

CALCULATOR CHECK: 2 [ENTER] 3 [×] 5 [×] [g] [FACT] → X = 30

## PRIME BLOCK HEIGHT REFERENCE

| PRIME | LOG10(P) | VISIBLE HEIGHT | UNITS (U) | COLOR |
|---|---|---|---|---|
| [ 2 ] | 0.3010 | 7.00 mm | 1.00 U | Blue |
| [ 3 ] | 0.4771 | 11.09 mm | 1.58 U | Gold |
| [ 5 ] | 0.6990 | 16.25 mm | 2.32 U | White |
| [ 7 ] | 0.8451 | 19.65 mm | 2.81 U | Gray |
| [ 11 ] | 1.0414 | 24.21 mm | 3.46 U | Gold |
| [ 13 ] | 1.1139 | 25.90 mm | 3.70 U | White |
| [ 17 ] | 1.2304 | 28.61 mm | 4.09 U | Gray |
| [ 19 ] | 1.2788 | 29.73 mm | 4.25 U | Gold |
| [ 23 ] | 1.3617 | 31.67 mm | 4.52 U | Gray |
| Box Bay | Stepped | 84.00 mm | 12.0 U | White |
| Lid Tray | 12-Sock | 10.40 mm | 1.49 U | White |

### WHY YOUR BRAIN LOVES THIS

Physical heights replace rote memorization — your hands and eyes spot common factors and multiples before you even write an equation!

1:1 CAD REFERENCE VIEW




BOX ICON & FORMULA QUICK-REFERENCE

**Box Button Icon Key:**
[PRIME] / [g][FACT] Prime Factorize

**Tower Height Formula:**
$H_{vis}(N) = 23.25 \text{ mm} \times \log_{10}(N)$

**Multiply Two Towers:**
Stack A on B ↦ A [ENTER] B [×]

**Divide Tower A by B:**
Pop B off A ↦ A [ENTER] B [÷]

**GCD & LCM Keystrokes:**
A [ENTER] B [GCF] or [f] [LCM]

# EXAMPLE 1: HANDS-ON WALKTHROUGH

Build Prime Towers for 12 & 18 Side-by-Side on the Lid Tray



## 1 PRIME DECOMPOSITION & TOWER CONSTRUCTION

**GOAL: Find the Least Common Multiple of 12 and 18: LCM(12, 18)**

1. DECOMPOSE NUMBER 12 INTO PRIMES:
12 = 2 × 6 = 2 × 2 × 3 = $2^2 \times 3$
→ **Gather two [ 2 ] blocks and one [ 3 ] block.**

2. DECOMPOSE NUMBER 18 INTO PRIMES:
18 = 2 × 9 = 2 × 3 × 3 = $2 \times 3^2$
→ **Gather one [ 2 ] block and two [ 3 ] blocks.**

3. BUILD & SEAT BOTH TOWERS ON THE LID TRAY:
Snap the blocks into Tower 12 and Tower 18 and seat them on the Lid Tray.

### LID TRAY SIDE-BY-SIDE HEIGHT COMPARISON

Seat both towers into adjacent sockets on the 12-socket Lid Tray:

- **Tower 12 Height: Visible height = 7.00 + 7.00 + 11.09 = 25.09 mm.**
Built from two Blue [ 2 ] blocks and one Gold [ 3 ] block ($2^2 \times 3 = 12$).
- **Tower 18 Height: Visible height = 7.00 + 11.09 + 11.09 = 29.18 mm.**
Stands 4.09 mm taller on the lid tray because a [ 3 ] replaces a [ 2 ]!

✓ **Physical confirmation: Shared base [2]×[3] = 6 is visible across both towers.**

**STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR** [PRIME] / [g][FACT

**STEP 1 (FACTOR 12):**
Press 12 [g] [FACT] → Shows $2^2 \times 3$, matching two Blue [2]s + one Gold [3] (25.09 mm).

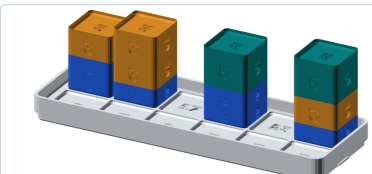
**STEP 2 (FACTOR 18):**
Press 18 [g] [FACT] → Shows $2 \times 3^2$, matching one Blue [2] + two Gold [3]s (29.18 mm).

**STEP 3 (FIND GCF & LCM):**
Press 12 [ENTER] 18 [GCF] (→ 6) and 12 [ENTER] 18 [f] [LCM] (→ 36) to verify both towers!

**LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):**
**T: 0 | Z: 0 | Y: 12 | X: 18 → DISPLAY (X) = 18**

## LESSON 1: BLOCKS ON LID TRAY ↔ [g][FACT] KEYSTROKES

**STEP A: HANDS-ON WITH BLOCKS ON LID TRAY** 1:1 CAD SCENE



| Tower 12 = [2][2][3] | Tower 18 = [2][3][3] | Tower 30 = [2][3][5] |
|---|---|---|
| 25.09 mm | 29.18 mm | 34.34 mm |

▲ **Seat prime towers in the 12-socket Lid Tray to compare** quantities and heights side-by-side!

**STEP B: TRY IT ON YOUR CALCULATOR** BOX ICON: [PRIME] / [g][FACT

FACTOR TREES: 12 = $2^2 \times 3$ vs 18 = $2 \times 3^2$




12 (25.09 mm) vs 18 (29.18 mm)

PRESS BOX ICON & RPN KEYS (SCG):

1. FACTOR 12:
1 2 g FACT → $2^2 \times 3$

2. FACTOR 18:
1 8 g FACT → $2 \times 3^2$

✓ DISPLAY (X) = 18

**KEYSTROKES: 12 [g] [FACT] 18 [g] [FACT]**
Build the factor towers on the Lid Tray, then press the Box Icon operation [g][FACT] to verify!

✓ **4-REGISTER STACK: T: 0 | Z: 0 | Y: 12 | X: 18 → X = 18**

# EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

Tower Leveling: Why GCD × LCM Always Equals A × B



## 2 PRIME UNION & THE CONSERVATION THEOREM

**1. FIND SHARED PRIMES (INTERSECTION):**
Tower 12 {2, 2, 3} and Tower 18 {2, 3, 3} share: { [ 2 ], [ 3 ] }
→ **Tower GCD = [ 2 ] + [ 3 ] = 6 (GCD(12, 18) = 6)**

**2. FIND PRIME UNION (HIGHEST POWERS):**
Take maximum of each prime: two [2]s and two [3]s
→ **Tower LCM = [ 2 ] + [ 2 ] + [ 3 ] + [ 3 ] = 36 (LCM(12, 18) = 36)**

### ✓ THEY ARE THE EXACT SAME HEIGHT!

**GCD(A, B) × LCM(A, B) = A × B**

**6 × 36 = 12 × 18 = 216**

Stack Tower 12 on top of Tower 18: [2]+[2]+[3] + [2]+[3]+[3] = 216.
Stack Tower GCD (6) on top of Tower LCM (36): [2]+[3] + [2]+[2]+[3]+[3] = 216.
**Both stacks contain exactly three [2]s and three [3]s!**

### LAW OF CONSERVATION OF PRIME MATTER

Primes are indestructible arithmetic matter. In finding GCD and LCM,
no blocks are created or destroyed — only partitioned and rearranged!

### ★ 2-PLAYER GAME: 'LID TRAY TOWER MATCH' (2 MINS)

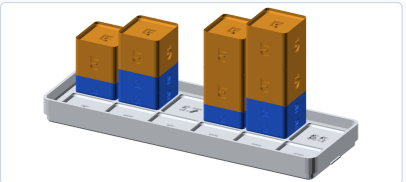
Player A builds a 3-block prime tower on the Lid Tray (like [2][3][5] = 30).
Player B must build a tower with the exact same height using a different block order ([5][3][2])
**and press [g][FACT] on the calculator to prove both equal 30!**

### ⚙ WHERE THIS SHOWS UP IN REAL-WORLD ENGINEERING & CRYPTOGRAPHY

Logarithmic addition (log(A×B) = log(A)+log(B)) is how engineers multiplied numbers for centuries!
Modern internet security (RSA encryption) relies on the fact that stacking two giant prime
**blocks is easy, while figuring out which primes built a giant tower is supercomputer-hard.**

## LESSON 2: GCD & LCM TOWERS ↔ [GCF] & [f][LCM] KEYS

STEP A: HANDS-ON WITH BLOCKS ON LID TRAY — 1:1 CAD SCENE

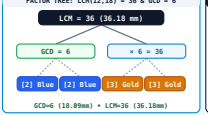


| GCD(12,18) = 6 | LCM(12,18) = 36 | 6 × 36 = 12 × 18 |
|---|---|---|
| 18.09 mm | 36.18 mm | 216 (54.27 mm) |

▲ On the Lid Tray, GCD = [2][3] (6) is the shared core and LCM = [2][2][3][3] (36) is the full union!

STEP B: TRY IT ON YOUR CALCULATOR — BOX ICON: [PRIME] / [g][FACT

FACTOR TREE: LCM(12,18) = 36 & GCD = 6




GCD=6 (18.09mm) • LCM=36 (36.18mm)

PRESS BOX ICON & RPN KEYS (SC6):

1. FIND GCF: 12 ENTER 18 GCF = 6
2. FIND LCM: 12 ENTER 18 f LCM = 36

✓ DISPLAY (X) = 36

KEYSTROKES: 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [f] [LCM]
Press [GCF] for the shared prime tower (6) and [f][LCM] for the full union tower (36).
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 6 | X: 36 → X = 36

# EXAMPLE 3 & DESK CHALLENGES

Commutative Towers: 2 × 3 × 5 = 30 + 3 Desk Challenges



SKU: PRIME-01 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

## 3 COMMUTATIVE TOWERS: 2 × 3 × 5 = 30 IN ANY ORDER

**THE GOAL: Prove that block order never changes tower height or product!**

**1. BUILD TOWER [2][3][5]:**

Stack Blue [2] (7.0mm) + Gold [3] (11.1mm) + White [5] (16.3mm) = 34.34 mm.

**2. BUILD REVERSED TOWER [5][3][2]:**

Stack White [5] + Gold [3] + Blue [2] on the Lid Tray in reverse order.

**3. COMPARE ON THE LID TRAY:**

Seat both towers in adjacent sockets—their summits stand dead-level at 34.34 mm!

### ✓ THE COMMUTATIVE LAW IN 3D: 2 × 3 × 5 = 30

**Every block's height is proportional to its logarithm: h(p) = K_log × log10(p).**

Since heights add linearly: h(2) + h(3) + h(5) = h(5) + h(3) + h(2) = 34.34 mm.

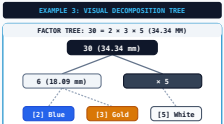
Order never changes the sum of heights or the product of factors (2 × 3 × 5 = 30).

Verify on calculator: 2 [ENTER] 3 [×] 5 [×] [g][FACT] proves both equal 30!

**PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER**

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

**EXAMPLE 3: VISUAL DECOMPOSITION TREE**




**STEP B: PRESS BOX ICON & RPN KEYS (SC6)**

TACTILE KEYSTROKE WALKTHROUGH:

1. STACK 2×3×5:




2. BOX ICON KEY:




✓ CALCULATOR DISPLAY (X) = 30

FULL KEYSTROKES: 2 [ENTER] 3 [×] 5 [×] [g] [FACT]

✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 30 → X = 30

## 3 DESK CHALLENGES — TRY THESE NOW!

**WARM-UP #1** **Build 8 and 12 on the Lid Tray. Find GCD & LCM!**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #1 below!

**LEVEL-UP #2** **Find the only 3-block tower that equals 30.**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #2 below!

**BRAIN-MELTER #3** **Build 20 and 50. Verify GCD × LCM = 20 × 50.**

Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator

✓ Build first, then flip booklet upside-down to check Answer #3 below!

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)

#1: GCD = 4, LCM = 40 (40 × 10 = 100 × 4 = 1,000)

#2: GCD = 10, LCM = 30 (2 × 3 × 5 = 30 (spheric number!)

#3: GCD = [2][2] = 4, LCM = [2][2][5] = 20 | 4 × 20 = 80

**FOR PARENTS & TEACHERS (Grades 4–9 (Number Theory, Factors, GCD & LCM))**

- CCSS.MATH.CONTENT.6.NS.B.4: Find greatest common factor & least common multiple.
- CCSS.MATH.CONTENT.4.OA.B.4: Find factor pairs & identify prime vs. composite numbers.
- TEKS 111.26.b.6.7.A: Generate equivalent expressions using prime factorization.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

**BOSS CHALLENGE #4** **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([PRIME] / [g][FACT]) and keystrokes on your calculator:

• 2 [ENTER] 3 [×] 5 [×] [g] [FACT] → X = 30

**CHECK OFF YOUR MASTERY MILESTONES:**

- ☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
- ☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
- ☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
- ☐ 4. **Packed all pieces back into the box in under 10 seconds (see Page 06 →)**

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: PACK-UP MAP → PAGE 06

# HOW TO STACK IT ALL BACK IN

The 10-Second Self-Checking Pack-Up Puzzle



## TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:

| | | | | | |
|---|---|---|---|---|---|
| [2]^11 "2048" | [2]^11 "2048" | [3]^7 "2187" | [3]^7 "2187" | [5]^4·3 "1875" | [5]^4·3 "1875" |
| [7]^4 "2401" | [11]^3·2 "2662" | [13]^3 "2197" | [17]^2·8 "2312" | [19]^2·6 "2166" | [23]^2·4 "2116" |

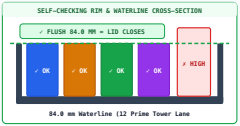
### ✓ THE 84.0 MM STORAGE BOX WATERLINE CHECK (▼)

Each of the 12 storage box columns has a stepped floor pedestal calibrated to its prime. When all 70 prime blocks are stacked in their home columns inside the Storage Box, all 12 towers rise to the exact same 84.00 mm waterline so the front door slides shut. If one column dips low, you know immediately which prime block is still on the desk!

1. STORAGE CASE & COMPARTMENTS



2. WHY THE LID SELF-CHECKS EVERY PIECE




▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

## THE 10-SECOND PACK-UP ROUTINE

### HOW TO STACK EVERYTHING BACK IN:

1. SORT BY COLOR & NUMBER:
Group matching blocks or tiles together on your desk first.

2. STACK & DROP INTO LABELED BAYS:
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!

3. TUCK THIS CARD IN THE LID & SNAP SHUT:
Fold this guide back to [01] and rest it right on top before closing.

### 12-SOCKET LID TRAY & SLIDING FRONT DOOR

- Twelve recessed sockets on the Lid Tray hold towers upright while you compare heights.
- Slide the front door into the box rails and snap the Lid Tray on top for backpack storage.
- Easy care: wipe clean or rinse in lukewarm soapy water (never put in dishwasher).

### SHORT A PIECE AT CLEAN-UP TIME?

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

### ✓ PRIME-01 4-POINT CLEAN-UP CHECKLIST

| | |
|---|---|
| 1. 84MM WATERLINE: | All 12 prime columns inside the box stand dead-level at 84.0 mm. |
| 2. CLEAR LID TRAY: | Remove any comparison towers still seated in the 12 Lid Tray sockets. |
| 3. FOLD CARD TO [01]: | Fold this reference booklet to Panel [01] and lay inside the lid. |
| 4. SLIDE DOOR & SNAP: | Slide the front retention door down and snap the Lid Tray shut! |

### MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)

Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2). All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

# THE STACKCALC CONNECTED LAB

How Your Kits & RPN Calculator Stack Work Together



## ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

## WHERE YOUR BOX FITS IN THE 9-KIT LAB

| | | |
|---|---|---|
| BASE-10<br>**Ten-Frames Lab**<br>Early Counting & Regrouping | DOMI-01<br>**Math Domino Studio**<br>Doubles & 12-Frame Part-Whole | FRAC-01<br>**Fraction Studio**<br>Proportional Towers & 4-Way Faces |
| PRIME-01 ◂ YOUR BOX<br>**Prime Factor Studio**<br>Multiply by Stacking Prime Towers | ALGB-01<br>**Algebra Studio**<br>Quadratic Area Arrays & Factoring | TIME-01<br>**Planetary Clock**<br>12:1 Geared Hands & Time Wedges |
| PROB-01<br>**Probability Dice**<br>Fair Polyhedrals & 10-Sector Spinner | VOLV-01<br>**Unit Circle Volvelle**<br>Spin-to-Read Exact Trig Radicals | SC-32-HW<br>**SC-32 RPN Calculator**<br>4-Register Stack Scientific Engine |

### WHY THIS MATTERS: ZERO "THROW-AWAY" MATH

In most classrooms, every grade uses a totally unrelated plastic toy.
In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. PRIME-01 BOX LID ICON



**[PRIME] / [g][FACT]**
**EMBOSSED ON BOX LID**
Build First → Press Key!

2. SC-6 ELEMENTARY RPN



3. SC-32 SCIENTIFIC RPN



▲ **BOX ICON = CALCULATOR KEY CONTINUITY:**
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

## HOW BLOCKS CONNECT TO THE CALCULATOR

### SHARED MODULAR FOOTPRINT & HEIGHTS

- Universal 7mm base unit: Prime Block [2] (7.0mm) matches the 1/12th Fraction block & :05 Clock block.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

### WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- Stacking blocks mirrors the SC-32 RPN calculator: [2] [ENTER] [3] [×] builds the tower digitally.
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

### FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

**Explore All 9 Kits & Free Guides at stackcalc.io**

**LIVE 4-REGISTER RPN STACK**

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 6 (GCF) (1st Block)
X: 36 (LCM) (Active Display)

**HOW BLOCKS MAP TO STACK KEYS**

1. **PLACE 1ST PIECE ON STAGE:**
Type number + [ENTER] → lifts to Y
2. **ADD / STACK 2ND PIECE:**
Type 2nd number in X, then [+]/[×]
3. **PRESS BOX ICON KEY:**
Press [PRIME] / [g][FACT]
✓ **Zero Parentheses • Always Visible!**

**BOX ICON = [PRIME] / [g][FACT] • VERIFIED LESSON SEQUENCES:**
- Lesson 1: 12 [g] [FACT] 18 [g] [FACT] → X = 18
- Lesson 2: 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [f] [LCM → X = 36

# KEEP THIS CARD! PART IDs & VIDEO

Complete 72-Piece Reference & 60-Second 3D Video Guide



## PART ID REFERENCE TABLE (72 TOTAL PIECES)

✓ **KEEP THIS CARD INSIDE YOUR BOX LID!**
Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

| PART NAME | QTY | COLOR / FINISH | PART ID |
|---|---|---|---|
| Prime Block [ 2 ] | 29x | Blue | PRIM-BLCK-02 |
| Prime Block [ 3 ] | 17x | Gold | PRIM-BLCK-03 |
| Prime Block [ 5 ] | 8x | White | PRIM-BLCK-05 |
| Prime Block [ 7 ] | 4x | Gray | PRIM-BLCK-07 |
| Prime Block [ 11 ] | 3x | Gold | PRIM-BLCK-11 |
| Prime Block [ 13 ] | 3x | White | PRIM-BLCK-13 |
| Prime Block [ 17 ] | 2x | Gray | PRIM-BLCK-17 |
| Prime Block [ 19 ] | 2x | Gold | PRIM-BLCK-19 |
| Prime Block [ 23 ] | 2x | Gray | PRIM-BLCK-23 |
| Storage Box, Door & Lid | 2x | White | GEN-PRIM-3bf |

**NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.**
Enter any Part ID above at **stackcalc.io/parts** to order individual replacement pieces.

**COLOR & PIECE IDENTIFICATION KEY** — MATCH ANY LOOSE PART

| Blue | Gold | White | Gray |
|---|---|---|---|
| Prime [ 2 ] (29x) | Primes [ 3, 11, 19 ] | Primes [ 5, 13 ], Box | Primes [ 7, 17, 23 ] |

**UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)**
Measure any loose piece in 7mm units (U) against this scale bar to identify it:




✓ **DISTRICT PO & SAFETY: 10-Pack SKU PRIME-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS**

## WATCH THE 60-SECOND 3D VIDEO GUIDE

**POINT CAMERA TO WATCH DEMO**
**stackcalc.io/products/prime-studio/**
See animated 3D examples & pack-up walkthrough

**QUESTIONS, CLASSROOM SETS & SUPPORT**
• Email us anytime: support@stackcalc.io
• School & District POs: school-orders@stackcalc.io
**Designed in Mountain View, CA • Non-Toxic & Washable**

**COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES**

**BOOK CHAPTERS:** SC-6 Guide Ch. 3 (Prime Towers) • SC-32 Manual Ch. 2
**STANDARDS:** CCSS 4.OA.B.4, 6.NS.B.4 • TEKS 5.4A, 6.7A (Primes, GCF & LCM)
**WATERLINE SPEC:** 84.0 mm Waterline (12 Prime Tower Lanes)

**ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC6):**
**Ex 1:** 12 [g] [FACT] 18 [g] [FACT] → **X=18**
**Ex 2:** 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [f] [LCM] → **X=36**
**Ex 3:** 2 [ENTER] 3 [×] 5 [×] [g] [FACT] → **X=30**

**BOX ICON KEY: [PRIME] / [g][FACT]**
✓ **Cross-Referenced to Companion Book & Calculator Engine**
Fold back to [01] & keep this reference card inside your box lid!

**LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts**

✓ **KEEP THIS GUIDE IN BOX LID!**